EEL5840 Fundamentals of Machine Learning **Fall 2024**
Midterm Exam **Solutions**

The make-up midterm exam for EEE4773 excluded question 6 below and had a different point distribution.

1. (10 points) **Suppose that you are training a linear regression model for a training set with N data points $\{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}$, and its corresponding target labels $\{t_i\}_{i=1}^N$. Consider the feature representation**

$$\phi(x_i) = \left[x_i^2, e^{x_i}, \sin(2\pi x_i), \cos(2\pi x_i), \log(1+x_i), \frac{1}{1+x_i}\right]^T$$

Answer the following questions:

(a) (3 points) **Write down the mapper function.**

The mapper function is defined as:

$$y(x_i) = \phi(x_i)^T \mathbf{w} + w_0$$

where $\mathbf{w} = [w_1, w_2, \dots, w_6]^T$

(b) (3 points) **Suppose you want to minimize the absolute error with the Lasso regularizer. Write down the objective function.**

The objective function is defined as:

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^{N} \left|t_i - (\phi(x_i)^T \mathbf{w} + w_0)\right| + \lambda \sum_{j=0}^{6} |w_j| \\ &= \sum_{i=1}^{N} |t_i - y(x_i)| + \lambda \sum_{j=0}^{6} |w_j| \\ &= \|\mathbf{t} - \mathbf{X}\mathbf{w}\|_1 + \lambda \|\mathbf{w}\|_1 \end{aligned}$$

(c) (4 points) **What is the Bayesian interpretation of this objective function? Show and justify your work.**

$$
\begin{aligned}
\arg_{\mathbf{w}} \min J(\mathbf{w}) &= \arg_{\mathbf{w}} \max -J(\mathbf{w}) \\
&= \arg_{\mathbf{w}} \max \exp\left(-J(\mathbf{w})\right) \\
&= \arg_{\mathbf{w}} \max \exp\left(-\sum_{i=1}^{N} |t_i - y(x_i)| - \lambda \sum_{j=0}^{6} |w_j|\right) \\
&= \arg_{\mathbf{w}} \max \prod_{i=1}^{N} \exp\left(-|t_i - y(x_i)|\right) \prod_{j=0}^{M} \exp\left(-\lambda |w_j|\right) \\
&\propto \arg_{\mathbf{w}} \max \prod_{i=1}^{N} \mathcal{L}(t_i|\phi(x_i)^T\mathbf{w}, 1) \prod_{j=0}^{M} \mathcal{L}(w_j|0, 1/\lambda) \\
&= \arg_{\mathbf{w}} \max \mathcal{L}(\mathbf{t}|\mathbf{X}\mathbf{w}, 1)\mathcal{L}(\mathbf{w}|0, 1/\lambda)
\end{aligned}
$$

As shown above, the Bayesian interpretation of this objective function shows that minimizing this objective function is equivalent to perform a parameter estimation with MAP, where the data likelihood on the target values is Laplacian-distributed, with a combined prior probability on the parameters $\mathbf{w}$.

2. (5 points) **What is the role of cross-validation in machine learning?**

   Cross-validation is a crucial technique in machine learning used to assess how model generalizes to an independent dataset (the validation set). Cross-validation is utilized within the training phase to help identify the set of best hyperparameter values to use and other model design choices.

3. (5 points) **Suppose that you are working with a two-dimensional features space, where x_1 and x_2 are the two features. Upon receiving a sample $\mathbf{x} = [x_1, x_2]^T$, the goal is to predict a continuous value t. Assume that you have examples of training pairs $\{(x_1, x_2)_i, t_i\}_{i=1}^{N}$. Suppose we want to train a quadratic mapper function of the form,**

$$f(x_1, x_2) = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + w_4(x_1)^2 + w_5(x_2)^2$$

   Provide a solution for $\mathbf{w} = [w_0, w_1, w_2, w_3, w_4, w_5]^T$.

   Instead of working directly in the two-dimensional feature space (x_1, x_2), we can build a feature mapping of the form

$$
\begin{aligned}
\phi : \mathbb{R}^2 &\longrightarrow \mathbb{R}^6 \\
(x_1, x_2) &\longmapsto (1, x_1, x_2, x_1x_2, x_1^2, x_2^2)
\end{aligned}
$$

   Let the feature matrix, $\mathbf{X}$, be an $N \times 6$ matrix that stores the 6-dimensional feature representation (including the dummy input 1), $\phi(\mathbf{x})$, for each sample $\mathbf{x}_i = (x_{i1}, x_{i2}), i = 1, \ldots, N$.

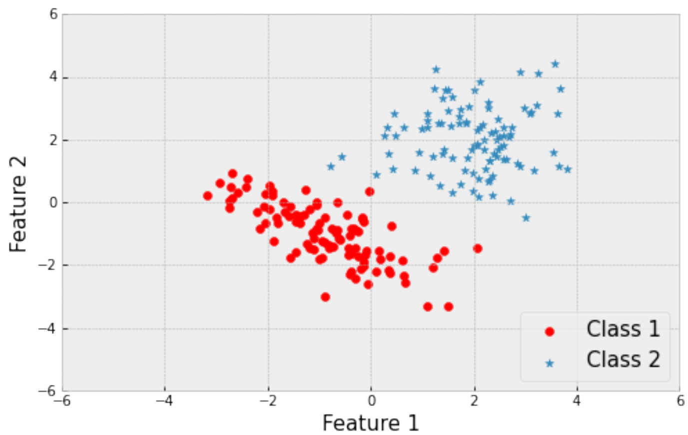
Now, we can write the mapper function:

$$y(\mathbf{x}) = \mathbf{X}\mathbf{w}$$

This is a simple linear regression problem. Using the least squares objective function, we know that the solution for the parameters $\mathbf{w}$ is:

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{t}$$

4. (10 points) **Consider the dataset depicted below composed of two classes (class 1 and class 2) in a 2-dimensional feature space. Each class has 100 samples. This is what the data looks like:**



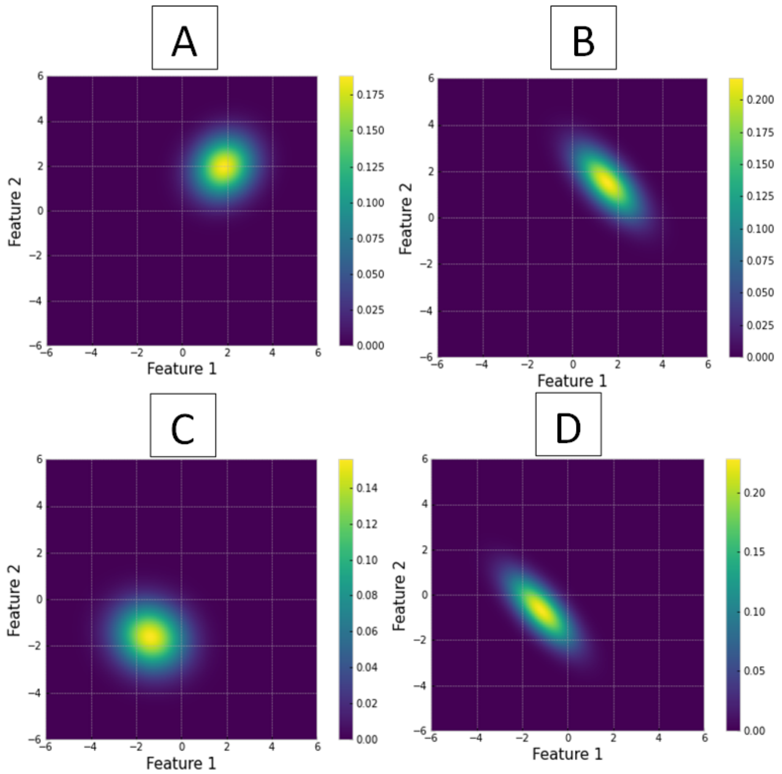
Answer the following questions:

(a) (3 points) **The four plots below (A, B, C and D) represent contours plots of the data likelihood as modeled with a Gaussian distribution. From the four plots, select two plots: one that represents the data likelihood for class 1, and another that represents the data likelihood for class 2. Justify your answer.**

- Class 1: the data has approximately a sample mean of $[-1, -1]$ and an elliptical covariance matrix with negative correlation between features 1 and 2. The best contour fitting this description is surface D.
- Class 2: the data has approximately a sample mean of $[2, 2]$ and a circular covariance matrix with approximately zero correlation between features 1 and 2. The best contour fitting this description is surface A.

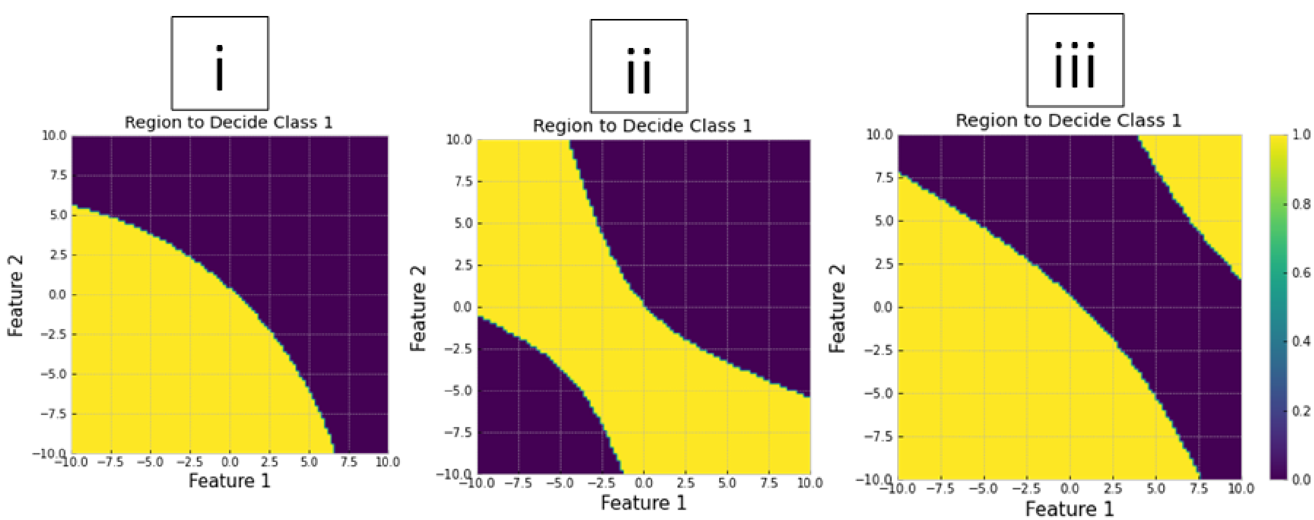
(b) (4 points) **In order to train the Naïve Bayes Classifier for this dataset, what other information do you need obtain in order to be able to make predictions? Provide your estimates below.**

In addition to the data likelihoods for class 1 and class 2, we also need to define the prior probability for each class. Namely, we can consider $P(C_k) = \frac{N_k}{N}$ where N_k is



the number of samples assigned to class C_k, N is the total number of samples and $k = \{1, 2\}$.

(c) (3 points) **The three plots below (i, ii and iii) represent the decision surface for deciding class 1 (red class). Based on data likelihoods you selected in part a, which of the following plots corresponds to the decision surface for deciding class 1? Justify your answer.**



Decision surface ii corresponds to the decision surface for deciding class 1, because (1) the sample mean is included in the region to decide class 1, and (2) the decision surface shows that the data likelihood covers quadrants 2 and 4 to accommodate the large uncertainty along that direction.

5. (20 points) **Suppose you have a training set with N data points $\{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}^+$ (set of positive real numbers). Assume the samples are independent and identically distributed (i.i.d.), and each sample is drawn from a Gamma random variable with probability density function:**

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

where $\alpha, \beta > 0$.

Moreover, consider another Gamma density as the prior probability on the hyperparameter β,

$$p(\beta|a,b) = \frac{b^a}{\Gamma(a)} \beta^{a-1} e^{-b\beta}$$

where $a, b > 0$.

Answer the following questions:

(a) (5 points) **Derive the maximum likelihood estimate (MLE) for the parameter β. Show your work.**

The observed data likelihood is:

$$\mathcal{L}^0 = \prod_{i=1}^N \frac{\beta^\alpha}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-\beta x_i}$$

The log-likelihood is given by:

$$\begin{aligned}\mathcal{L} &= \ln \mathcal{L}^0 \\ &= \sum_{i=1}^N \left(\alpha \ln(\beta) - \ln(\Gamma(\alpha)) + (\alpha - 1)\ln(x_i) - \beta x_i\right)\end{aligned}$$

We can now find the MLE estimation for β:

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \iff \sum_{i=1}^N \left(\alpha \frac{1}{\beta} - x_i\right) = 0 \iff \frac{\alpha N}{\beta} = \sum_{i=1}^N x_i \iff \beta = \frac{\alpha N}{\sum_{i=1}^N x_i}$$

(b) (5 points) **Derive the maximum a posteriori (MAP) estimate for the parameter β. show your work.**

The observed data likelihood for MAP is:

$$
\begin{aligned}
\mathcal{L}^0 &= \left(\prod_{i=1}^{N} \frac{\beta^\alpha}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-\beta x_i}\right) \frac{b^a}{\Gamma(a)} \beta^{a-1} e^{-b\beta} \\
&\propto \left(\prod_{i=1}^{N} \beta^\alpha e^{-\beta x_i}\right) \beta^{a-1} e^{-b\beta} \\
&= \beta^{\sum_{i=1}^N \alpha} e^{-\beta \sum_{i=1}^N x_i} \beta^{a-1} e^{-b\beta} \\
&= \beta^{N\alpha+a-1} e^{-\beta\left(\sum_{i=1}^N x_i + b\right)}
\end{aligned}
$$

We can now find the MAP estimation for β:

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \beta} = 0 &\iff (N\alpha + a - 1)\ln(\beta) - \beta\left(\sum_{i=1}^{N} x_i + b\right) = 0 \\
&\iff \beta = \frac{N\alpha + a - 1}{\sum_{i=1}^N x_i + b}
\end{aligned}
$$

(c) (5 points) **Is the Gamma distribution a conjugate prior for the parameter β, of the Gamma data likelihood distribution? Why or why not?**

Yes, the Gamma distribution forms a conjugate prior because the shape of the posterior probability is proportionally equal to the prior probability.

(d) (5 points) **Suppose you would like to update the Gamma prior distribution and the MAP point estimation in an online fashion, as you obtain more data. Write the pseudo-code for the online update of the prior parameters. In your answer, specify the new values for the parameters of the prior.**

The pseudo-code for online update of the prior is as follows:

1. Start at iteration $t = 0$. Initialize the prior parameters $a^{(t)}$ and $b^{(t)}$.
2. Compute the parameter estimation for the current prior probability:

$$\beta_{\text{MAP}} = \frac{N\alpha + a^{(t)} - 1}{\sum_{i=1}^N x_i + b^{(t)}}$$

3. Update the prior parameters

$$
\begin{aligned}
a^{(t+1)} &\leftarrow a^{(t)} + N\alpha \\
b^{(t+1)} &\leftarrow b^{(t)} + \sum_{i=1}^{N} x_i
\end{aligned}
$$

4. Increment iteration counter

$$t \leftarrow t + 1$$

6. (5 points) **Let $\theta_{\mathbf{MLE}}$ and $\theta_{\mathbf{MAP}}$ be the parameter estimates using MLE and MAP, respectively. In what scenarios do we achieve $\theta_{\mathbf{MLE}} = \theta_{\mathbf{MAP}}$?**

   The parameter estimate for MLE and MAP will be the same if: (1) the MAP uses an uninformative prior, like the uniform distribution; or (2) sample size is very large.

7. (25 points) **Consider a training set containing positive real numbers ($x \in \mathbb{R}^+$) for 2 classes, C_0 and C_1. The training set has 400 samples for class C_0 and 100 for C_1.**

   Suppose that you have reason to believe that samples belonging from C_0 are drawn from an Exponential random variable with parameter $\lambda > 0$, and samples belonging to C_1 are drawn from a Gamma random variable with parameters $\alpha > 0$ and $\beta > 0$. In other words:

$$p(x|C_0) = \lambda e^{-\lambda x}$$
$$p(x|C_1) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

   where $\Gamma(x) = (x-1)!$. Answer the following questions:

   (a) (5 points) **For a given training set $\{(x_i, t_i)\}_{i=1}^N$ with $x_i \in \mathbb{R}^+$ and $t_i \in \{0, 1\}$, how would you determine the parameters for each classes' data likelihood (λ, α and β)? Explain in words a list of steps for finding the point estimates. (No derivations.)**

   The parameters λ, α and β can be estimated as point estimates of the data likelihood either using MLE, a completely data-driven approach without any regularization of the parameter values; or using MAP, the Maximum A Posteriori approach which will induce a constraint on the parameter values.

   (b) (5 points) **Can you regularize your point estimates for λ, α and β? If yes, clearly explain how you would proceed. If not, explain why not.**

   Yes, as mentioned in the previous question, λ, α and β can be regularized. In order to do this, we would have to use the MAP approach.

   (c) (10 points) **For next two parts, consider $\lambda = 2$, $\alpha = 2$ and $\beta = 1$. Consider the test point $x = 3$. Which class (C_0 or C_1) will it be assigned to? Show your work.**

   The prior probability for each class can be estimated as its relative frequency in the training set, namely

$$p(C_1) = \frac{400}{500} = \frac{4}{5} \text{ and } p(C_2) = \frac{100}{500} = \frac{1}{5}$$

   For the provided parameter values, the data likelihood for class 1 and class 2 are defined as:

$$p(x|C_0) = 2e^{-2x}$$
$$p(x|C_1) = \frac{1^2}{\Gamma(2)}x^{2-1}e^{-1x} = xe^{-x}$$

For the test point $x = 3$, we compute the posterior probability with

$$P(C_k|x=3) = \frac{p(x=3|C_k)p(C_k)}{p(x=3)} = \frac{p(x=3|C_k)p(C_k)}{\sum_{j=1}^{2} p(x=3|C_j)p(C_j)}$$

We now find:

$$P(C_0|x=3) = \frac{2e^{-6}\frac{4}{5}}{2e^{-6}\frac{4}{5} + 3e^{-3}\frac{1}{5}} \approx 0.117$$
$$P(C_1|x=3) = \frac{3e^{-3}\frac{1}{5}}{2e^{-6}\frac{4}{5} + 3e^{-3}\frac{1}{5}} \approx 0.883$$

Since $P(C_1|x=3) > P(C_0|x=3)$ then the test point $x = 3$ is assigned to class C_1.

(d) (5 points) **For a given test sample x, provide an equation that will determine all cases in each x will be assigned to C_0. Show your work.**

A point x is assigned to class C_0 if $P(C_0|x) > P(C_1|x)$:

$$P(C_0|x) > P(C_1|x)$$
$$\frac{p(x|C_0)p(C_0)}{p(x)} > \frac{p(x|C_1)p(C_1)}{p(x)}$$
$$\frac{p(x|C_0)}{p(x|C_1)} > \frac{p(C_1)}{p(C_0)}$$
$$\frac{2e^{-2x}}{xe^{-x}} > \frac{1/5}{4/5}$$
$$\frac{e^{-x}}{x} > \frac{1}{8}$$

$\therefore x \in C_0$ if $\frac{e^{-x}}{x} > \frac{1}{8}$.

8. (20 points) **Consider the dataset $\mathbf{X} = \{x_i\}_{i=1}^{N}$, where $x_i \geq 0, \forall i$. Suppose your goal is to perform density estimation using a Mixture Model, in particular, a Rayleigh Mixture Model. Its data likelihood can be written as:**

$$f(x) = \sum_{k=1}^{K} \pi_k g_k\left(x|\sigma_k\right)$$

where

$$\sum_{k=1}^{K} \pi_k = 1$$

and

$$g_k\left(x|\sigma_k\right) = \frac{x}{\sigma_k^2} e^{-x^2/(2\sigma_k^2)}$$

with $\sigma_k > 0$ and $0 \leq \pi_k \leq 1, \forall k$. Answer the following questions:

(a) (2 points) **Assuming your data is i.i.d., write down the observed data likelihood, $\mathcal{L}^0$.**

The observed data likelihood is:

$$\mathcal{L}^0 = \prod_{i=1}^{N} \sum_{k=1}^{K} \pi_k \frac{x_i}{\sigma_k^2} e^{-x_i^2/(2\sigma_k^2)}$$

(b) (2 points) **Use the Expectation-Maximization (EM) algorithm to find the maximum likelihood solutions. Start by introducing the hidden latent variables Z. Describe precisely what they are.**

Yes, we can introduce the hidden latent variable Z, where z_i corresponds to the Rayleigh component from which sample x_i was drawn from. Moreover, since we have a total of K Rayleigh components, $z_i \in \{1, 2, \ldots, K\}$.

(c) (3 points) **For the hidden variables you defined above, write down the complete data likelihood, $\mathcal{L}^c$.**

The complete data likelihood is given by:

$$\mathcal{L}^c = \prod_{i=1}^{N} \pi_{z_i} \frac{x_i}{\sigma_{z_i}^2} e^{-x_i^2/(2\sigma_{z_i}^2)}$$

(d) (4 points) **Write down the EM optimization function, $Q(\Theta, \Theta^t)$, where $\Theta = \{\pi_k, \sigma_k\}_{k=1}^{K}$. Your final solution should contain the sum of simple log-terms.**

The EM optimization function is given by:

$$
\begin{aligned}
Q(\Theta, \Theta^t) &= \mathbb{E}_z[\ln(\mathbf{L}^c)|\mathbf{X}, \Theta^t] \\
&= \sum_{z_i=1}^{K} \ln(\mathbf{L}^c) P(z_i|x_i, \Theta^t) \\
&= \sum_{k=1}^{K} \left[\sum_{i=1}^{N} \left(\ln(\pi_k) + \ln(x_i) - 2\ln(\sigma_k) - \frac{x_i^2}{2\sigma_k^2} \right) \right] P(z_i = k|x_i, \Theta^t) \\
&= \sum_{k=1}^{K} \left[\sum_{i=1}^{N} \left(\ln(\pi_k) + \ln(x_i) - 2\ln(\sigma_k) - \frac{x_i^2}{2\sigma_k^2} \right) \right] C_{ik}
\end{aligned}
$$

(e) (4 points) **Derive the update equations for the parameters σ_k.**
The solution for the parameter σ_k is:

$$
\begin{aligned}
\frac{\partial Q(\Theta, \Theta^t)}{\partial \sigma_k} &= 0 \\
\sum_{i=1}^{N} \left(-\frac{2}{\sigma_k} + \frac{4\sigma_k x_i^2}{(2\sigma_k^2)^2} \right) C_{ik} &= 0 \\
\sum_{i=1}^{N} \left(-2\sigma_k^2 + x_i^2 \right) C_{ik} &= 0 \\
\sum_{i=1}^{N} 2\sigma_k^2 C_{ik} &= \sum_{i=1}^{N} x_i^2 C_{ik} \\
\sigma_k &= \sqrt{\frac{\sum_{i=1}^{N} x_i^2 C_{ik}}{\sum_{i=1}^{N} 2C_{ik}}}
\end{aligned}
$$

(f) (5 points) **Derive the update equations for the parameters π_k.**
In order to find the solution for π_k, we must add the constraint $\sum_{k=1}^{K} \pi_k = 1$ to the optimization function:

$$
Q_\pi(\Theta, \Theta^t) = Q(\Theta, \Theta^t) - \lambda \left(\sum_{k=1}^{N} \pi_k - 1 \right)
$$

The solution for π_k is then:

$$\frac{\partial Q_\pi(\Theta, \Theta^t)}{\partial \pi_k} = 0$$
$$\sum_{i=1}^{N} \frac{1}{\pi_k} C_{ik} - \lambda = 0$$
$$\pi_k = \frac{1}{\lambda} \sum_{i=1}^{N} C_{ik}$$

Since $\sum_{k=1}^{K} \pi_k = 1$, we find that: $\sum_{k=1}^{K} \frac{1}{\lambda} \sum_{i=1}^{N} C_{ik} = 1 \iff \lambda = N$. Hence, the solution for π_k is:

$$\pi_k = \frac{\sum_{i=1}^{N} C_{ik}}{N}$$